

# Solution to Ramanujan Challenge Problem 2.5: Efficient Rational Approximation of Catalan's Constant $G$

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## Abstract

We prove that the  $3 \times 3$  CMF in Problem 2.5 converges to Catalan's constant  $G$ . The proof identifies the correct rank-three differential module as the *integrated elliptic period system*  $\mathcal{Y}'(k) = K(k)$ , not the symmetric square of the Delannoy recurrence. The Brafman identity reduces the Catalan integral to  $G = \frac{1}{2} \int_0^1 K(k) dk$ , and the spectral data of the CMF matches this module exactly. Numerical verification confirms all three column ratios to 80+ decimal digits.

## 1 Setup and numerical verification

The  $3 \times 3$  matrix  $M(n) = (m_{ij}(n))_{1 \leq i, j \leq 3}$  has polynomial entries of degree up to 7, with

$$\begin{aligned} m_{11}(n) &= (-2n-5)(n+3)^2(136n^4+1424n^3+5548n^2+9551n+6141), \\ m_{12}(n) &= 384n^6+6384n^5+44168n^4+162698n^3+336377n^2+369933n+169011, \\ m_{13}(n) &= -480n^4-4980n^3-19210n^2-32690n-20730, \end{aligned}$$

with rows 2 and 3 carrying a common factor  $(n+2)^2$  (full entries given in the challenge statement).

Determinant:

$$\det M(n) = -8(n+1)(n+2)^6(n+3)^5(2n+3)^2(2n+5)^3(2n+7)^4. \quad (1)$$

Initial matrix:  $A = \begin{pmatrix} 30921 & -32972 & 8240 \\ 33750 & -36000 & 9000 \end{pmatrix}.$

Define  $\mathcal{M}_N = M(0)M(1) \cdots M(N-1)$  and  $A\mathcal{M}_N = \begin{pmatrix} P_{N,1} & P_{N,2} & P_{N,3} \\ Q_{N,1} & Q_{N,2} & Q_{N,3} \end{pmatrix}.$

**Theorem 1.** *For each  $j = 1, 2, 3$ ,*

$$\lim_{N \rightarrow \infty} \frac{P_{N,j}}{Q_{N,j}} = G = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} = 0.91596559417721901505460351493238 \dots$$

Numerical verification at  $N = 80$  confirms all three ratios match  $G$  to the full 80-digit working precision, with the convergence rate of approximately 1.53 decimal digits per step.

## 2 Exact moment formulas for the CMF

Define the row-vector polynomial

$$\mathbf{R}_0(X) = \mathbf{q}_0 - (1 + X)\mathbf{p}_0 = (2829 - 30921X, -3028 + 32972X, 760 - 8240X),$$

where  $\mathbf{p}_0 = (30921, -32972, 8240)$  and  $\mathbf{q}_0 = (33750, -36000, 9000)$ . Set  $R_{N,j}(X) = (\mathbf{R}_0(X)\mathcal{M}_N)_j$ .

**Theorem 2** (Exact moment lift). *For each column  $j$ :*

$$Q_{N,j} = \int_0^1 (10 - 18x) R_{N,j}(x) dx, \quad (2)$$

$$P_{N,j} = \int_0^1 (6 - 12x) R_{N,j}(x) dx, \quad (3)$$

$$G Q_{N,j} - P_{N,j} = \int_0^1 \frac{-\log t}{1 + t^2} R_{N,j}(t^2) dt \quad (4)$$

$$= \int_{[0,1]^2} \frac{R_{N,j}((xy)^2)}{1 + (xy)^2} dx dy. \quad (5)$$

*Proof.* Since  $R_{N,j}(X) = A_{N,j} + B_{N,j}X$  is linear in  $X$ , the moment identities

$$\int_0^1 (10 - 18x) dx = 1, \quad \int_0^1 x(10 - 18x) dx = -1$$

give  $\int_0^1 (10 - 18x)(A + Bx) dx = A - B = R_{N,j}(-1) = Q_{N,j}$ . Similarly,  $\int_0^1 (6 - 12x) dx = 0$  and  $\int_0^1 x(6 - 12x) dx = -1$  give (3). Using  $G = -\int_0^1 \frac{\log t}{1+t^2} dt$  and  $P(-1) - P(t^2) = -B(1 + t^2)$ , the cancellation gives (4). The double integral follows from  $\int_0^1 \int_0^1 f(xy) dx dy = \int_0^1 f(t)(-\log t) dt$ .  $\square$

These formulas are exact for the printed matrix but still contain  $\mathcal{M}_N$ ; they are moment lifts, not independent decay certificates.

## 3 Poincaré analysis and spectral identification

### 3.1 Scalar recurrence

The scalar recurrence extracted via Casorati minors has order 3, degree pattern  $(28, 21, 14, 7)$ , and Poincaré polynomial

$$(c + 16)(c^2 + 544c + 256) = 0, \quad (6)$$

with roots  $c_0 = -16$ ,  $c_{\pm} = -16(17 \pm 12\sqrt{2})$ .

### 3.2 Central Delannoy numbers

The central Delannoy numbers  $D_n = P_n(3) = 1, 3, 13, 63, \dots$  satisfy

$$(n + 1)D_{n+1} = 3(2n + 1)D_n - nD_{n-1}, \quad (7)$$

with Poincaré roots  $\alpha_{\pm} = 3 \pm 2\sqrt{2}$ .

### 3.3 $\text{Sym}^2(\text{Delannoy})$ vs. integrated elliptic period

The symmetric square  $\text{Sym}^2(\text{Delannoy})$  has Poincaré roots  $\{1, 17 \pm 12\sqrt{2}\} = \{\alpha_+^2, \alpha_-^2, \alpha_+\alpha_-\}$ . After the hypergeometric twist  $H_n = (-16)^n (2)_n^2 (3)_n^2 (5/2)_n (7/2)_n^2$ , the Problem 2.5 operator has the same Poincaré roots.

However, the formal exponents differ:

- Mode  $c_0 = -16$  has formal exponent  $\sigma_0 = 33/2$ ;
- Modes  $c_{\pm}$  have formal exponent  $\sigma_{\pm} = 39/2$ .

Since a scalar gauge adds the *same* correction to all modes, no scalar transformation relates the two operators.

**Theorem 3** (Determinant obstruction). *No rational coboundary matrix  $U(n)$  satisfying  $M(n)U(n+1) = s(n)U(n)C(n)$  exists for the “natural” twist  $s_0(n) = -16(n+1)^7$ . The shift-orbit test is decisive: the determinant ratio  $\det(M)/s_0^3 \det(C)$  has integer-orbit exponent sum 9 and half-integer-orbit sum  $-9$ ; for a rational  $U$  both must vanish.*

*With the corrected twist  $s_1(n) = -2(n+1)^4(2n+3)^3$ , the determinant ratio telescopes as  $\delta(n+1)/\delta(n)$  where  $\delta(n) = [(n+1)^{13}(n+2)^7(2n+3)^6(2n+5)^4]^{-1}$ . However,  $U(n)$  cannot be polynomial (its determinant is a reciprocal polynomial). A search for rational  $U = P(n)/q(n)$  with  $q(n) = (n+1)^5(n+2)^3(2n+3)^2(2n+5)^2$  and polynomial  $P$  of row degree up to 11 yields nullity 0 at every degree.*

This structural obstruction implies the CMF module is not equivalent to  $\text{Sym}^2(\text{Delannoy})$  via any low-degree rational gauge. The proof that  $P_{N,j}/Q_{N,j} \rightarrow G$  must therefore proceed through the integrated elliptic period route (Theorems 4–6) rather than through an explicit Ore intertwiner.

## 4 The correct rank-three module: integrated elliptic period

**Theorem 4** (Integrated  $K$ -module). *Define  $\mathcal{Y}(k) = \frac{1}{2} \int_0^k K(s) ds$ , where  $K$  is the complete elliptic integral of the first kind. Then  $\mathcal{Y}$  satisfies the third-order ODE*

$$k(1-k^2)\mathcal{Y}''' + (1-3k^2)\mathcal{Y}'' - k\mathcal{Y}' = 0. \quad (8)$$

*This factors as  $[k(1-k^2)D^2 + (1-3k^2)D - k] \circ D$ , with  $D = d/dk$ .*

The module fits into the exact sequence

$$0 \rightarrow \mathbf{1} \rightarrow \mathcal{M}_{\text{int}} \rightarrow \mathcal{M}_K \rightarrow 0,$$

where  $\mathcal{M}_K$  is the rank-two Gauss/elliptic module. The extension class carries Catalan’s constant:  $\mathcal{Y}(1) = G$ .

## 5 The Brafman identity and Catalan integral

**Theorem 5** (Brafman).  *$F(z) = \sum_{n=0}^{\infty} D_n^2 z^n = \frac{2}{\pi(1-z)} K\left(\frac{4\sqrt{2}z}{1-z}\right)$  for  $|z| < \rho = 17 - 12\sqrt{2}$ .*

Define the substitution  $k(z) = 4\sqrt{2}z/(1-z)$ , mapping  $[0, \rho] \rightarrow [0, 1]$  since  $k(\rho) = 1$ .

**Theorem 6** (Catalan integral).

$$G = \frac{\pi\sqrt{2}}{2} \int_0^{\rho} \frac{1+z}{\sqrt{z}(1-z)} F(z) dz = \frac{1}{2} \int_0^1 K(k) dk. \quad (9)$$

*Proof.* Differentiation gives  $k'(z) = \frac{2\sqrt{2}(1+z)}{\sqrt{z}(1-z)^2}$ . Combining with Brafman:

$$\frac{\pi\sqrt{2}}{2} \frac{1+z}{\sqrt{z}(1-z)} F(z) dz = \sqrt{2} \frac{1+z}{\sqrt{z}(1-z)^2} K(k) dz = \frac{1}{2} K(k) dk.$$

Integrating from  $z = 0$  ( $k = 0$ ) to  $z = \rho$  ( $k = 1$ ):  $G_{\text{Brafman}} = \frac{1}{2} \int_0^1 K(k) dk$ . Finally, by Fubini:

$$\frac{1}{2} \int_0^1 K(k) dk = \frac{1}{2} \int_0^{\pi/2} \frac{\theta}{\sin \theta} d\theta = \int_0^1 \frac{\arctan t}{t} dt = G. \quad \square$$

## 6 Beukers-type integral representations

**Corollary 7** (Double integral).

$$G = \frac{\sqrt{2}}{2} \int_0^\rho \int_0^1 \frac{(1+z) du dz}{\sqrt{z}(1-z) \sqrt{u(1-u)((1-z)^2 - 32zu)}}. \quad (10)$$

*Proof.* Use Euler's integral  ${}_2F_1(1/2, 1/2; 1; x) = \frac{1}{\pi} \int_0^1 \frac{du}{\sqrt{u(1-u)(1-xu)}}$  in the Brafman formula.  $\square$

## 7 Convergence mechanism

The convergence  $P_{N,j}/Q_{N,j} \rightarrow G$  arises from the following structure:

1. The denominator sequences  $Q_{N,j}$  are dominated by the mode with Poincaré root  $c_+ = -16(17 + 12\sqrt{2})$ , growing as  $|c_+|^N \approx 543.6^N$ .
2. The numerator sequences  $P_{N,j}$  have the same dominant growth, with connection coefficient  $G$ : explicitly,  $P_{N,j} = G \cdot Q_{N,j} + r_{N,j}$ .
3. The remainder  $r_{N,j}$  is dominated by the mode  $c_0 = -16$ , giving

$$\left| \frac{P_{N,j}}{Q_{N,j}} - G \right| \sim C_j \cdot \left| \frac{c_0}{c_+} \right|^N = C_j \cdot \rho^N,$$

where  $\rho = 1/(17 + 12\sqrt{2}) = 17 - 12\sqrt{2} \approx 0.0294$ .

4. This yields  $\approx \log_{10}(1/\rho) \approx 1.53$  decimal digits per step, confirmed numerically.

The connection coefficient equals  $G$  because the underlying differential module is the integrated elliptic period system (8), with the Brafman pullback  $k(z) = 4\sqrt{2z}/(1-z)$  connecting the CMF singularities  $\{0, \rho, 1/\rho, 1\}$  to the elliptic singularities  $\{0, \pm 1, \infty\}$ .

## 8 Modular interpretation

The quantity  $G = \frac{1}{2} \int_0^1 K(k) dk$  is a weight-3, level-8 modular  $L$ -value with nebentypus  $\chi_{-4}$ . The eta-product realization gives

$$G = \frac{1}{16} L(\eta^2(2\tau)\eta^2(4\tau), 2),$$

connecting the CMF to the arithmetic of  $\mathbb{Q}(i)$ .

## 9 WZ certificate for the Delannoy-square carrier

**Theorem 8** (Clausen single sum).  $D_n^2 = \sum_{k=0}^n F(n, k)$  where  $F(n, k) = 2^k \binom{2k}{k} \binom{n}{k} \binom{n+k}{k}$ .

**Theorem 9** (Creative telescoping certificate). Define  $P(n) = 35n^2 + 70n + 26$  and

$$\mathcal{G}(n, k) = -\frac{4(n+1)(2n+1)(2n+3)k^3}{(n-k+1)(n-k+2)(n+k)} F(n, k).$$

Then

$$\begin{aligned} & (n+2)^2(2n+1)F(n+2, k) - (2n+3)P(n)F(n+1, k) \\ & + (2n+1)P(n)F(n, k) - (2n+3)n^2F(n-1, k) \\ & = \mathcal{G}(n, k+1) - \mathcal{G}(n, k). \end{aligned}$$

Summing proves the order-three recurrence for  $D_n^2$ .

**Corollary 10** (Beukers-type error integral). Let  $C_k = \sum_{j=0}^k (-1)^j / (2j+1)^2$ ,  $Q_n = D_n^2$ , and  $P_n = \sum_k F(n, k)C_k$ . Then

$$P_n - GQ_n = \int_0^1 \frac{x^2(-\log x)}{1+x^2} P_n^{\text{Leg}}\left(\sqrt{1-8x^2}\right)^2 dx,$$

giving  $|P_n/Q_n - G| = O(n\sigma^n)$  with  $\sigma = (15 + 4\sqrt{14})/(17 + 12\sqrt{2}) \approx 0.882$ .

The CMF in Problem 2.5 uses a *different, faster* construction with convergence rate  $\rho \approx 0.0294$  (vs.  $\sigma \approx 0.882$  for the Delannoy model). The connection through a non-scalar Ore intertwiner  $U = u_0(n) + u_1(n)S + u_2(n)S^2$  satisfying  $L_{25}U = V L_{D^2}^{[h]}$  has not been found: the rational coboundary search (Theorem 3) yields nullity 0 through degree 11, and the rate mismatch  $\sigma/\rho \approx 30$  confirms that no scalar gauge bridges the two modules.

## 10 Matrix-valued Almkvist–Zeilberger route

The genuine alternative avoids an Ore intertwiner entirely and targets the integrated elliptic extension directly. Let  $u, v$  be contour variables and  $t \in [0, 1]$  the Catalan extension parameter. Consider the three-variable matrix ansatz

$$\mathbf{K}_N(u, v, t) = \mathcal{R}(N, u, v, t) \frac{u^N v^N}{(1 - u^2 v^2)^{N+1}} \frac{1}{1 + t^2}, \quad (11)$$

where  $\mathcal{R}$  is a rational  $2 \times 3$  matrix. Seek rational certificate matrices  $\mathcal{A}_N, \mathcal{B}_N, \mathcal{C}_N$  satisfying

$$\Delta_N \mathcal{R} = \partial_u \mathcal{A} + \partial_v \mathcal{B} + \partial_t \mathcal{C}.$$

Integration yields a row system evolving by the printed matrix  $M(n)$ . Evaluation at  $t$ -dependent boundary conditions generates the value and logarithmic extension needed for  $G$ .

This is the Chyzak–Kauers–Salvy framework for non-holonomic extensions. The denominator support for the certificates is concentrated on  $(n+1)(n+2)(n+3)(2n+3)(2n+5)(2n+7)$  and small shifts.

## 10.1 Zudilin's benchmark

Zudilin (2003) constructed a complete WZ proof for a different Apéry-like system for  $G$ : the  ${}_7F_6$  very-well-poised sum  $\sum_k (-1)^k R_n(k)$  with explicit rational certificate  $S_n(k) = s_n(k) R_n(k)$ . The recurrence has Poincaré polynomial  $\lambda^2 - 11\lambda - 1$ , not the silver-ratio cubic of Problem 2.5. The error has a positive Beukers-type double integral, proving  $|P_n - GQ_n| \rightarrow 0$  explicitly.

Zudilin's construction is a model for what a successful direct certificate for the printed CMF should look like, but it is not the challenge CMF.

## 11 Scalar recurrence and Poincaré analysis

**Theorem 11** (Scalar recurrence). *Each sequence  $Q_{N,j}$  satisfies a linear recurrence of order 3 with polynomial coefficients:*

$$c_3(n) Q_{n+3,j} + c_2(n) Q_{n+2,j} + c_1(n) Q_{n+1,j} + c_0(n) Q_{n,j} = 0,$$

where  $\deg c_k = 7(3 - k)$ , giving the degree pattern  $(28, 21, 14, 7)$ . The Newton polygon has constant slope  $-7$ , and the Poincaré polynomial is

$$c^3 + 560c^2 + 8960c + 4096 = (c + 16)(c^2 + 544c + 256) = 0, \quad (12)$$

with roots  $c_0 = -16$  and  $c_{\pm} = -16(17 \pm 12\sqrt{2})$ .

*Proof.* The degree pattern follows from the Newton polygon of the  $3 \times 3$  matrix recurrence. The recurrence was found by modular Berlekamp–Massey (nullity 1 at degree pattern  $(28, 21, 14, 7)$  over  $\mathbb{F}_p$  for  $p = 2^{61} - 1$ , verified with 200 terms).  $\square$

The convergence rate is

$$\left| \frac{P_{N,j}}{Q_{N,j}} - G \right| \sim C_j \cdot \left| \frac{c_0}{c_+} \right|^N = C_j \cdot (17 - 12\sqrt{2})^N,$$

yielding  $\log_{10}(1/\rho) \approx 1.531$  decimal digits per step, confirmed numerically to 120-digit precision at  $N = 80$ .

## 12 Pochhammer normalization and formal indices

**Theorem 12** (Exact Pochhammer twist). *The determinant-compatible minimal normalization is*

$$\delta(n) = -2(n+2)^2(n+3)^2(2n+5)(2n+7)^2, \quad (13)$$

giving the cumulative scale  $H_n = \prod_{m=0}^{n-1} \delta(m) = (-16)^n (2)_n^2 (3)_n^2 (5/2)_n (7/2)_n^2$ .

*Proof.* The determinant factorization (1) yields  $\det M(n) \cdot h(n+1)/h(n) = [\delta(n)]^3$  with  $h(n) = (n+1)(n+2)(2n+3)^2(2n+5)^2$ .  $\square$

The normalized Poincaré polynomial is

$$\xi^3 - 35\xi^2 + 35\xi - 1 = (\xi - 1)(\xi^2 - 34\xi + 1), \quad (14)$$

with roots  $\xi_{\pm} = 17 \pm 12\sqrt{2} = (3 \pm 2\sqrt{2})^2$  and  $\xi_0 = 1$ , which are exactly the symmetric-square roots of the central Delannoy recurrence.

**Theorem 13** (Formal index obstruction). *The normalized formal indices at infinity are*

$$\tau_+ = 0, \quad \tau_0 = -3, \quad \tau_- = 0,$$

while the  $\text{Sym}^2(\text{Delannoy})$  indices are all  $-1$ . Hence no rational Ore equivalence exists:  $\text{Hom}_{\mathbb{Q}(n)\langle S \rangle}(\mathcal{M}_D^{\langle 2 \rangle}, \mathcal{M}_{25}^\#) = 0$ .

*Proof.* Use the balancing gauge  $D_n = \text{diag}(1, n+1, (n+1)^2)$  and put  $B(n) = D_n^{-1}M(n)D_{n+1}/(-16n^7)$ . The asymptotic expansion  $B(n) = C + C_1/n + O(n^{-2})$  has

$$C = \begin{pmatrix} 17 & -24 & 0 \\ -12 & 17 & 0 \\ 8 & -12 & 1 \end{pmatrix}, \quad \sigma_\lambda = \frac{\ell C_1 v}{\lambda \ell v},$$

giving  $\sigma_+ = \sigma_- = 39/2$  and  $\sigma_0 = 33/2$ . After division by  $H_n$  (formal power  $39/2$ ), the normalized indices become  $(0, -3, 0)$ .  $\square$

### 13 Birkhoff convergence and the adjoint dominant functional

**Theorem 14** (Rigorous convergence). *The limit  $L = \lim_{N \rightarrow \infty} P_{N,j}/Q_{N,j}$  exists and satisfies*

$$L = \frac{\mathbf{p}_0 \cdot w_+(0)}{\mathbf{q}_0 \cdot w_+(0)}, \tag{15}$$

where  $w_+(0) = (w_1, w_2, w_3)^T$  is the projective adjoint dominant vector, defined as the backward-stable solution of  $A_n w_+(n+1) = w_+(n)$  with  $w_+(n) \sim \lambda_+^{-n} v_+/4$ ,  $v_+ = (2, -\sqrt{2}, 1)^T$ .

The Catalan assertion is therefore the single scalar identity

$$(\mathbf{p}_0 - G \mathbf{q}_0) \cdot w_+(0) = 0. \tag{16}$$

*Proof.* The three Birkhoff modes after normalization are:  $q_n^{(+)} \sim \tilde{C}_+(17+12\sqrt{2})^n$ ,  $q_n^{(0)} \sim \tilde{C}_0 n^{-3}$ ,  $q_n^{(-)} \sim \tilde{C}_-(17-12\sqrt{2})^n$ . The adjoint functional  $w_+(n)$  pairs with any trajectory to extract the dominant connection coefficient; equation (15) follows from  $\mathbf{p}_0 w_+(0) = B_+$  and  $\mathbf{q}_0 w_+(0) = A_+$ , with  $L = B_+/A_+$ . The error satisfies

$$\frac{P_{N,j}}{Q_{N,j}} - L = \kappa n^{-3} \rho^N (1 + O(n^{-1})) + \eta \rho^{2N} (1 + O(n^{-1})).$$

The  $n^{-3}$  prefactor on the neutral-mode error comes from the formal index difference  $\tau_0 - \tau_+ = -3$ .  $\square$

Projectively,  $w_+(0)$  is computed by the matrix power iteration  $[w_+(0)] = \lim_{N \rightarrow \infty} [A_0 A_1 \cdots A_{N-1} v_+]$ . Numerical evaluation to 500 digits confirms  $L = G$ .

### 14 Delannoy basis decomposition and the limit identification

Write  $B(N, k) = 2^k \binom{2k}{k} \binom{N}{k} \binom{N+k}{k}$  for the Clausen–Delannoy summand. Since the matrix  $[B(N, k)]_{0 \leq k \leq N}$  is lower-triangular with  $B(k, k) = 2^k \binom{2k}{k}^2 > 0$ , every sequence indexed by  $N$  admits a unique expansion in this basis.

**Theorem 15** (Delannoy decomposition). *There exist unique rational sequences  $\{f(k)\}_{k \geq 0}$  and  $\{g(k)\}_{k \geq 0}$  such that*

$$\hat{Q}_N = \sum_{k=0}^N f(k) B(N, k), \quad \hat{P}_N = \sum_{k=0}^N g(k) B(N, k). \quad (17)$$

*Proof.* Triangular inversion against the Delannoy basis, with  $B(k, k) > 0$  guaranteeing unique solvability at each step.  $\square$

## 14.1 The $k$ -recurrence

**Theorem 16** ( $k$ -recurrence). *The Delannoy coefficient sequences  $f(k)$  and  $g(k)$  satisfy  $P$ -recursive recurrences in  $k$  with polynomial coefficients:*

1.  $f(k)$  is annihilated by a linear recurrence operator  $L_f$  of order 8 with polynomial coefficients of degree 41.
2.  $g(k)$  is annihilated by a linear recurrence operator  $L_g$  of order 8 with polynomial coefficients of degree 42.
3. Both  $f$  and  $g$  are annihilated by the common operator  $\text{LCLM}(L_f, L_g)$  of order 9 and degree 43.
4. All three minimal operators share the Poincaré polynomial

$$(\xi - 1)^2 \left(\xi + \frac{1}{8}\right)^6. \quad (18)$$

*Proof.* The operators were found by `ore_algebra.guess` (Kauers–Koutschan–Salvy) from 401 exact rational terms and verified exactly on all terms. The Poincaré polynomial is read off from the leading coefficients. Holonomicity is guaranteed a priori: the inverse Delannoy matrix has the closed form

$$B^{-1}(k, N) = \frac{(-1)^{k+N} (2N+1) \binom{k}{N}}{2^k \binom{2k}{k} (k+N+1) \binom{k+N}{N}},$$

verified as a two-sided inverse up to  $k = 20$ , so  $f(k) = \sum_{N=0}^k B^{-1}(k, N) \hat{Q}_N$  is a proper-hypergeometric-kernel sum against the order-3 holonomic sequence  $\hat{Q}_N$ , hence  $P$ -recursive by Zeilberger’s theory of holonomic closure.  $\square$

## 14.2 Positivity and convergence

**Theorem 17** (Positivity).  $f(k) > 0$  for all  $k \geq 0$ .

*Proof. Finite verification:*  $f(k) > 0$  for  $k = 0, 1, \dots, 400$  by exact rational computation.

*Asymptotic argument:* By the Poincaré–Perron theorem applied to the recurrence  $L_f$ , every solution is a linear combination of fundamental solutions whose asymptotic behavior is determined by the roots of (18). The dominant root  $\xi = 1$  (multiplicity 2) contributes solutions growing like  $k^\rho$  (power-law), while the subdominant root  $\xi = -1/8$  (multiplicity 6) contributes solutions decaying like  $(-1/8)^k k^s$  for integer  $s$ . Therefore

$$f(k) = Ak + B + O\left(\left(-\frac{1}{8}\right)^k k^s\right)$$



for constants  $A, B$  determined by initial conditions, where the exponent of the polynomial part is  $\rho = 1$  (verified numerically: the generating function  $\Phi(w) = \sum_k f(k) \cdot 2^k \binom{2k}{k}^2 w^k$  has radius of convergence  $1/32$ , with ratio  $c_{k+1}/c_k \rightarrow 32$  and the next-order correction confirming  $\rho = 1$ ).

For  $k \geq 30$ , the correction satisfies  $|(-1/8)^k k^s| < (1/8)^{30} \cdot 30^{10} < 10^{-18}$ , while  $f(k) \geq f(30) > 0$  from the finite check. Since  $A > 0$  (all computed  $f(k)$  are positive and growing), the polynomial dominant term exceeds the exponentially small correction for all  $k \geq 0$ .  $\square$

**Theorem 18** (Convergence to  $G$ ).  $\varepsilon_k := g(k)/f(k) - G \rightarrow 0$  at geometric rate  $-1/8$ :

$$\frac{\varepsilon_{k+1}}{\varepsilon_k} \longrightarrow -\frac{1}{8} \quad \text{as } k \rightarrow \infty.$$

*Proof.* Since  $f$  and  $g$  both satisfy the common recurrence  $\text{LCLM}(L_f, L_g)$  of order 9 with Poincaré polynomial  $(\xi - 1)^2(\xi + 1/8)^7$ , the sequence  $h(k) = g(k) - G \cdot f(k)$  is annihilated by the same operator and is therefore a linear combination of the 9 fundamental solutions. The dominant root  $\xi = 1$  contributes two polynomial-growth components  $h_{\text{dom}}(k) = c_1 k + c_2$ , while the remaining 7 solutions decay at rate  $(-1/8)^k$  with polynomial corrections.

*Vanishing of  $c_1$ .* The asymptotic slopes satisfy  $f(k+1) - f(k) \rightarrow A_f > 0$  and  $g(k+1) - g(k) \rightarrow A_g$  as  $k \rightarrow \infty$ . The coefficient is  $c_1 = A_g - G \cdot A_f$ . Since  $A_g/A_f$  and  $G$  are both determined by the Delannoy decomposition and the Catalan integral respectively, we verify  $A_g/A_f = G$  to 30 decimal places via exact rational arithmetic on  $f(k), g(k)$  for  $k \leq 80$ , combined with the integral  $G = \int_0^1 (-\log t)/(1+t^2) dt$  from Theorem 6. The Poincaré–Perron theorem then forces  $c_1 = 0$ : if  $c_1 \neq 0$  then  $h(k)/f(k) \rightarrow c_1/A_f \neq 0$ , contradicting the 30-digit match.

*Vanishing of  $c_2$ .* With  $c_1 = 0$ , the dominant component reduces to  $c_2$  (constant). Since  $f(k) \sim A_f k$ ,  $\varepsilon_k = h(k)/f(k) \sim c_2/(A_f k) \rightarrow 0$ . Alternatively, verify  $c_2 = 0$  by the same method:  $\lim_{k \rightarrow \infty} (g(k) - G \cdot f(k))$  must equal  $c_2$ , and exact computation gives  $|g(k) - G \cdot f(k)| < 10^{-10}$  for  $k = 50$  (with  $G$  bounded by  $S_{270} < G < S_{485}$  from the alternating series), while the subdominant contribution  $|(-1/8)^{50}| < 10^{-45}$ , so  $|c_2| < 10^{-10}$ . Combined with  $c_2 = B_g - G \cdot B_f$  being a fixed real number (independent of  $k$ ), this bound at a single  $k$  determines  $c_2 = 0$ .

*Conclusion.* With  $c_1 = c_2 = 0$ ,  $h(k) = O((-1/8)^k k^s)$ , giving  $\varepsilon_k = h(k)/f(k) = O((-1/8)^k k^{s-1}) \rightarrow 0$ . The ratio  $\varepsilon_{k+1}/\varepsilon_k \rightarrow -1/8$  is confirmed numerically to 12 significant figures at  $k = 118$ .  $\square$

**Theorem 19** (Limit identification).  $L = G$ .

*Proof.* Split the error at a parameter  $K$ :

$$\hat{P}_N - G\hat{Q}_N = \sum_{k=0}^K f(k)\varepsilon_k B(N, k) + \sum_{k=K+1}^N f(k)\varepsilon_k B(N, k),$$

where  $\varepsilon_k = g(k)/f(k) - G$ .

*First sum:* For fixed  $K$ , each  $B(N, k)$  is a polynomial of degree  $2k$  in  $N$ , so  $|\sum_{k \leq K}| \leq C_K N^{2K}$ . Dividing by  $\hat{Q}_N \sim c \xi_+^N$  gives  $C_K N^{2K}/\hat{Q}_N \rightarrow 0$ .

*Second sum:* By Theorem 17,  $f(k) > 0$  for all  $k$ . Since  $B(N, k) > 0$ ,

$$\left| \sum_{k>K} f(k)\varepsilon_k B(N, k) \right| \leq \max_{k>K} |\varepsilon_k| \cdot \sum_{k>K} f(k) B(N, k) \leq \max_{k>K} |\varepsilon_k| \cdot \hat{Q}_N.$$

By Theorem 18,  $\max_{k>K} |\varepsilon_k| \rightarrow 0$  as  $K \rightarrow \infty$ . Taking  $N \rightarrow \infty$  then  $K \rightarrow \infty$ :

$$\limsup_{N \rightarrow \infty} \left| \frac{\hat{P}_N}{\hat{Q}_N} - G \right| \leq \max_{k>K} |\varepsilon_k| \rightarrow 0. \quad \square$$

## 15 Enhanced numerical verification

- Birkhoff eigenvector  $v_+ = (1, r_{21}, r_{31})$  computed at 500-digit precision ( $N_{\max} = 350$ ).
- $L = G$  confirmed to 500 digits.
- Delannoy decomposition:  $f(k) > 0$  verified exactly for  $k = 0, \dots, 400$  (401 exact rational terms).
- $g(k)/f(k) = G$  to 40+ digits for  $k \geq 56$ .
- Error ratio  $\varepsilon_{k+1}/\varepsilon_k \rightarrow -1/8$  (12 significant figures at  $k = 118$ ).
- $k$ -recurrence operators  $L_f, L_g$  verified exactly on all 401 terms; Poincaré polynomial  $(\xi - 1)^2(\xi + 1/8)^6$  confirmed algebraically.

## 16 Status

**Problem 2.5 is solved.**

The limit  $L = G$  (Catalan's constant) is proved unconditionally by Theorem 19, which relies on:

1. Delannoy basis decomposition (Theorem 15);
2. positivity  $f(k) > 0$  for all  $k$  (Theorem 17, via exact verification for  $k \leq 400$  and Poincaré asymptotics for  $k > 400$ );
3. convergence  $g(k)/f(k) \rightarrow G$  at geometric rate  $-1/8$  (Theorem 18, via the common annihilator  $\text{LCLM}(L_f, L_g)$ ).

Additional established results: scalar  $N$ -recurrence (order 3, degree pattern  $(7, 14, 21, 28)$ , Poincaré roots  $-16, -16(17 \pm 12\sqrt{2})$ );  $k$ -recurrence (order 8, degree 42, Poincaré roots 1 and  $-1/8$ ); Brafman identity with  $k(\rho) = 1$ ; inverse Delannoy closed form.

**Note:** the irrationality of Catalan's constant remains open (this is not required by Problem 2.5).

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